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OSTP and NITRD NCO RFI
Development of an Artificial Intelligence (AI) Action Plan
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Request for Information Response

Development of an Artificial Intelligence (AI) Action Plan

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Introduction

Artificial intelligence (AI) is at the core of national security, economic competitiveness, and technological innovation. As the United States seeks to maintain its leadership in AI, it must leverage its free markets and entrepreneurial spirit to develop AI solutions that are effective and free from bias. The foundation of AI innovation is sound government policies that remove barriers, provide strategic direction, and ensure the flexibility needed to lead in a rapidly evolving technological landscape; two such critical policy areas are building a strong AI workforce and optimizing AI procurement processes. Without a well-trained and strategically placed workforce, AI innovations will struggle to transition from research to impactful government and industry applications. Likewise, without an efficient procurement system, the government risks duplicating investments, slowing adoption, and failing to fully leverage existing technological advancements.

Summary of Key Policy Recommendations

To ensure that the United States remains at the forefront of AI development and deployment, we propose the following concrete policy actions:

1. Education & Workforce – Strengthening AI Talent Pipelines

- a. **Increase AI workforce recruitment resources:** Expand federal initiatives to attract and retain top AI talent, including partnerships with universities, industry, and workforce development programs.
- b. **Expand AI hiring programs and pathways:** Develop structured hiring mechanisms, like direct-hire authorities and rotational AI fellowships across government agencies.
- c. **Create AI Talent Bridge Programs Between Universities and Government Agencies** – Establish structured transition pathways for students and early-career professionals to enter government AI roles through fellowships and direct hiring pipelines.
- d. **Improve AI education and training opportunities:** Invest in AI-focused curricula, apprenticeships, and upskilling programs to equip workers with the skills needed to succeed in an AI-driven economy.
- e. **Leverage data-driven workforce planning:** Implement standardized data collection and AI-based analytics to track hiring trends, workforce retention, and effectiveness of recruitment efforts.
- f. **Expand AI K-12 Education Initiatives and Early Talent Pipelines** – Invest in AI-focused STEM education, competitions, and mentorship programs to build long-term workforce readiness.
- g. **Support AI Public-Private Workforce Partnerships** – Encourage collaboration between federal agencies, industry leaders, and academia to co-develop AI training programs and career placement initiatives.



2. Procurement – Enhancing AI Research Coordination and Acquisition Efficiency

- a. **Streamline AI procurement processes:** Leverage existing (i.e Tradewinds tradewindai.com) fast-track procurement pathways for AI solutions that align with innovation priorities.
- b. **Strengthen AI technology transition mechanisms:** Improve collaboration between research institutions, private organizations, and operational units to accelerate the deployment of AI solutions in government operations.
- c. **Mandate AI procurement performance metrics:** Require agencies to track AI adoption rates, cost savings, and impact assessments to inform future funding priorities.
- d. **Expand AI-Specific Small Business Innovation Research (SBIR) and Small Business Technology Transfer (STTR) Grants** – Increase funding for small businesses developing AI solutions, ensuring broader participation in federal AI initiatives.
- e. **Create an AI Commercialization Accelerator for Small Businesses** – Establish a federally funded program to help small businesses transition AI research into deployable government solutions.
- f. **Create a centralized AI R&D knowledge repository:** Develop or leverage already developed platforms to track AI-related R&D investments and prevent redundant funding efforts.

Our Expertise

Mantel is a defense technology company focused on delivering solutions in the form of products and services to the National Security and dual-use markets. We provide solutions in Decision Support, Performance, and Safety and Survivability to enable advancements in human, canine and mission systems and achieve objectives in interoperability and organizational transformation. By integrating cutting-edge analytics, scalable open architectures, and mission-driven solutions, Mantel delivers impactful technologies that enhance operational readiness, efficiency, and decision-making for the U.S. defense sector. In addition to our technology capabilities, Mantel plays a critical role in defense workforce development through AI Futures, our dedicated program for civilian recruitment and engagement within the Department of Defense (DoD) as well as targeted geography specific efforts for local economic development. Our organization brings direct experience in AI workforce development and procurement innovation:

- **AI Futures** – A workforce initiative that identifies, recruits, and retains AI talent through sustained student engagement, strategic partnerships, and direct support to hiring organizations.



- **TruffleHog** – A procurement intelligence tool with advanced machine learning that enables federal agencies to identify, connect with, and leverage previously funded research and development (R&D) efforts.

Policy Focus Area 1: Education & Workforce

Artificial intelligence is reshaping industries, national security, and government operations, making a highly skilled AI workforce a national priority. However, the U.S. currently faces significant challenges in sourcing, recruiting, and retaining top AI talent, particularly within government agencies. Federal agencies struggle to attract AI professionals due to inefficient hiring processes, limited recruitment resources, and a lack of cohesive workforce planning. Additionally, the development of AI systems must be guided by technical excellence and national security priorities, free from ideological bias. To secure America's AI leadership, we must implement proactive workforce policies that create sustainable talent pipelines, streamline hiring, and ensure AI professionals can address future challenges.

The Problem: Gaps in AI Workforce Development

Despite AI's growing importance, the federal government faces structural and systemic hurdles in hiring and retaining AI professionals. The following challenges must be addressed:

1. Under-Resourced AI Hiring and Recruitment

- a. According to a National Association of Manufacturing and Deloitte report, the US will have a shortage of up to 3.5 million STEM workers by 2025.¹ Additionally, 1 in 2 AI jobs could be left unfulfilled by 2027.²
- b. Federal hiring efforts are not keeping pace with AI demand, with public-sector recruitment vastly outnumbered by private industry initiatives.

2. Decentralized and Inefficient Hiring Processes

- a. AI hiring lacks coordination across agencies, leading to redundant recruitment efforts, inconsistent skill requirements, and fragmented talent pipelines.
- b. Federal hiring processes are too slow to compete with the private sector, where AI roles are filled in an average of 30 days, compared to 100+ days in government.³

3. Competitive Disadvantages

- a. Many top AI professionals prefer industry jobs due to higher salaries and faster hiring cycles, leaving the government struggling to compete.
- b. The private sector is twice as likely to include any educational requirements for technical AI jobs, whereas the public sector relies more on associates and other standard degree requirements.⁴

¹ [The STEM skills gap: a growing challenge for the US to overcome](#)

² [Widening talent gap threatens executives' AI ambitions.](#)

³ [Modernizing federal hiring: Cutting bureaucracy, enhancing transparency, and strengthening the workforce](#)

⁴ [AI in the US Public Sector vs. Private Sector](#)



4. Lack of Standardized AI Workforce Data and Metrics

- a. There is no unified federal AI workforce database, making it difficult to assess hiring trends, skills gaps, and future workforce needs.
- b. AI workforce planning is largely reactive, with agencies struggling to forecast demand and implement proactive recruitment strategies.

Policy Recommendations: Strengthening AI Talent Pipelines

To address these challenges, we propose a series of policy actions to expand AI hiring programs, increase recruitment resources, and leverage workforce data to optimize hiring and retention.

1. Increase AI Workforce Recruitment Resources:

- a. **Scale federal AI recruitment personnel** – Increase the number of dedicated AI-focused recruiters in support of government agencies, ensuring AI hiring is prioritized alongside other high-demand technical fields.
- b. **Invest in nationwide AI recruitment campaigns** – Leverage social media, targeted advertising, university outreach, and virtual engagement strategies to connect AI professionals with federal career opportunities.
- c. **Establish AI recruitment incentive programs** – Offer signing bonuses, relocation assistance, student loan forgiveness, and competitive benefits packages to attract AI talent into government service.

2. Expand AI Hiring Programs and Pathways:

- a. **Create AI-focused direct-hire authorities** – Simplify hiring processes for AI professionals by allowing agencies to recruit and onboard AI talent quickly without lengthy competitive hiring cycles.
- b. **Launch rotational AI fellowships** – Establish structured programs that allow AI professionals to rotate between federal agencies, industry, and academia, facilitating cross-sector knowledge sharing and career development.
- c. **Expand AI internship-to-hiring pipelines** – Increase the number of paid AI internships and cooperative education programs, ensuring students and early-career professionals have clear transition paths into full-time government AI roles.

3. Create AI Talent Bridge Programs Between Universities and Government Agencies:

- a. **Establish formal AI fellowship and traineeship programs** – Develop structured partnerships with universities and industry to provide hands-on research and project-based learning experiences that prepare students for AI careers in government.
- b. **Develop AI hiring agreements with universities** – Create direct hiring partnerships with academic institutions, allowing government agencies to recruit AI talent directly from university programs.



- c. **Enhance AI career mentorship programs** – Implement government-sponsored mentorship initiatives that connect students and early-career professionals with experienced AI practitioners in federal agencies.
- 4. **Improve AI Education and Training Opportunities:**
 - a. **Invest in AI-focused curricula and degree programs** – Provide funding and incentives for universities, technical schools, and industry to develop AI-specific courses, certifications, and degree pathways that align with federal AI workforce needs.
 - b. **Expand AI apprenticeships and work-based learning programs** – Develop hands-on initiatives that integrate AI education with real-world applications in federal agencies and industry.
 - c. **Support AI upskilling and reskilling initiatives** – Fund programs that allow existing federal employees and private-sector workers to transition into AI-related careers through targeted training and certification programs.
- 5. **Leverage Data-Driven Workforce Planning:**
 - a. **Mandate standardized AI workforce data collection** – Require all federal agencies to track AI-related job openings, hiring trends, retention rates, and skills gaps to inform future workforce planning.
 - b. **Use AI-driven analytics to improve hiring strategies** – Leverage AI-powered tools to analyze workforce trends, predict future talent needs, and develop proactive recruitment strategies.
 - c. **Create an AI talent pipeline** – Develop a system that enables agencies to identify available AI professionals, reducing hiring bottlenecks and facilitating talent-sharing across agencies.
- 6. **Expand AI K-12 Education Initiatives and Early Talent Pipelines:**
 - a. **Invest in AI-focused STEM education programs** – Collaborate with industry to integrate AI concepts into K-12 curricula, ensuring early exposure to AI principles and career pathways.
 - b. **Support national AI competitions and challenges** – Fund AI hackathons, robotics competitions, and coding challenges to spark student interest and engagement in AI careers.
 - c. **Establish AI mentorship programs for K-12 students** – Connect AI professionals with students through virtual and in-person mentoring initiatives, particularly in underserved and rural communities.
- 7. **Support AI Public-Private Workforce Partnerships:**
 - a. **Develop AI training partnerships with industry and academia** – Facilitate cross-sector collaboration to ensure AI training programs align with real-world skills and workforce needs.



- b. Create AI workforce exchange programs** – Allow AI professionals to gain experience across federal agencies, private-sector companies, and research institutions through structured job rotation programs.
- c. Encourage industry-led AI workforce development initiatives** – Provide funding and policy support for private-sector companies to create AI training programs that align with government hiring needs.

Expected Impact of These Policies

Implementing these policy recommendations will yield significant long-term benefits, including:

- **Increased AI hiring efficiency** – By streamlining hiring processes and expanding recruitment efforts, agencies can reduce hiring timelines, ensuring critical AI positions are filled faster and with top-tier talent. This will enable federal agencies to stay ahead of rapid AI advancements rather than playing catch-up.
- **Stronger AI talent pipelines** – Investing in AI workforce development initiatives will create a sustainable pipeline of skilled professionals prepared to enter government roles. Through structured education-to-career pathways, apprenticeships, and direct hiring agreements, the government can consistently attract new AI talent and prevent workforce shortages.
- **Improved retention rates** – Offering competitive incentives, career development programs, and flexible AI hiring models will increase job satisfaction and reduce turnover among AI professionals in government. Clear career progression, mentorship programs, and ongoing upskilling opportunities will help retain top AI talent and prevent costly brain drain to the private sector.
- **Enhanced national security, economic growth, and innovation** – Strengthening AI workforce policies ensures the U.S. can maintain its leadership in AI-driven technological advancements, national security applications, and economic competitiveness. A robust AI workforce will enable faster adoption of AI solutions in defense, healthcare, infrastructure, and cybersecurity, driving progress across all sectors.

By investing in comprehensive, scalable, and strategically designed AI workforce development policies, we can future-proof America's AI leadership, ensuring long-term sustainability, global competitiveness, and a workforce capable of tackling the next generation of technological challenges.

Policy Focus Area 2: Procurement

As AI continues to expand and be at the forefront of national security, economic growth, and government modernization, efficient and strategic AI procurement is imperative. Despite investing billions in AI research and development (R&D), the federal government struggles with fragmented procurement efforts, limited cross-agency coordination, and slow technology



adoption, resulting in duplicative spending, inefficiencies, and missed opportunities to leverage existing advancements.

To maintain global AI leadership, the federal government must modernize its procurement infrastructure, enhance interagency collaboration, and leverage decision-making tools that ensure faster, more cost-effective, and strategically aligned technology transitions. This section identifies key challenges in AI procurement and presents actionable policy recommendations to streamline acquisition processes, optimize R&D investments, and accelerate the deployment of cutting-edge AI solutions across government agencies.

The Problem: Inefficiencies in AI Procurement and Transition

Despite significant federal investments in AI, procurement inefficiencies and fragmented acquisition strategies hinder technology adoption. The following challenges must be addressed:

1. Lack of Centralized Knowledge of AI R&D Investments

- a. A Government Accountability Office (GAO) report revealed that of 20 federal agencies, only five provided comprehensive information on their AI use cases. The remaining 15 had incomplete or inaccurate inventories, hindering effective oversight and coordination of AI initiatives.⁵
- b. The same GAO report found that agencies' inventories often contained data gaps and inaccuracies, making it challenging to identify and leverage existing AI solutions across departments.

2. Slow and Cumbersome AI Procurement Processes

- a. Federal procurement processes are often slow and inefficient, particularly when acquiring advanced technologies. These delays can lead to scenarios where technology becomes outdated before it is fully implemented, undermining the effectiveness of public sector operations.⁶
- b. Government agencies face unique challenges, including strict regulatory frameworks and compliance requirements, which slow down decision-making and technology adoption.⁷

3. Poor AI Technology Transition from R&D to Operations

- a. AI solutions often remain confined to pilot programs without clear pathways to full-scale implementation, limiting their operational impact.
- b. Complex federal acquisition rules pose significant hurdles for small businesses and startups, preventing innovative AI technologies from being integrated into government operations.

4. No Standardized Metrics for AI Procurement Success

⁵ [Most agency AI inventories are 'not fully comprehensive and accurate,' GAO reports](#)

⁶ [AI-Powered Procurement: Harnessing AI's Potential for More Efficient State Procurement Practices](#)

⁷ [Federal Agencies and Technology Adoption: The Struggle to Keep Pace with Innovation](#)



- a. Agencies lack data-driven procurement performance tracking, making it difficult to assess the effectiveness of AI investments.
- b. Without standardized return-on-investment metrics, decision-makers struggle to prioritize AI projects that offer the most significant benefits, leading to potential misallocation of resources.

Policy Recommendations: Optimizing AI Procurement & Technology Transition

To address these challenges, we propose concrete policy actions that will enhance AI procurement efficiency, reduce waste, and accelerate the transition of AI solutions from research to deployment.

1. Streamline AI Procurement Processes

- a. **Establish an AI Fast-Track Acquisition Program** – Create an expedited procurement framework for AI solutions that meet critical national security, public sector, and innovation priorities, reducing acquisition timelines and bureaucratic hurdles.
- b. **Enable AI Procurement as a Service (AIaaS)** – Promote flexible subscription-based procurement models that allow agencies to pilot, test, and scale AI technologies without committing to large, long-term contracts upfront.

2. Strengthen AI Technology Transition Mechanisms

- a. **Increase Cross-Sector AI Collaboration** – Foster partnerships between government agencies, private industry, and research institutions to identify promising AI technologies and develop structured pathways for transitioning them into government use.
- b. **Implement AI Transition Pilot Funds** – Establish dedicated funding mechanisms that provide agencies with the necessary resources to scale AI solutions beyond the pilot phase and into full operational deployment.
- c. **Develop AI Transition Roadmaps** – Require agencies to create clear implementation plans that outline milestones, testing phases, risk mitigation strategies, and agency adoption metrics for AI solutions.

3. Mandate AI Procurement Performance Metrics

- a. **Require Standardized AI Adoption Metrics** – Establish common performance indicators such as AI utilization rates, automation efficiency improvements, and impact on cost savings.
- b. **Implement AI Procurement ROI Assessments** – Develop a federal-wide framework for assessing the return on investment (ROI) and operational effectiveness of AI acquisitions.

4. Expand AI-Specific Small Business Innovation Research (SBIR) and Small Business Technology Transfer (STTR) Grants



- a. **Increase AI Grant Funding Allocations** – Expand SBIR/STTR funding pools to support AI startups, small businesses, and research initiatives, ensuring diverse participation in federal AI programs.
 - b. **Streamline the SBIR/STTR Application Process** – Reduce administrative burdens and improve funding accessibility by simplifying grant application requirements, review timelines, and eligibility criteria for AI-focused small businesses.
5. **Create an AI Commercialization Accelerator for Small Businesses**
 - a. **Provide Business Development and Technical Assistance** – Offer mentorship, market access, and commercialization training to AI startups and small businesses, helping them navigate federal procurement and successfully scale their technologies.
 - b. **Create AI Testbeds and Regulatory Sandboxes** – Establish controlled environments where small businesses can test and refine AI solutions in collaboration with federal agencies before full-scale adoption.
6. **Create a Centralized AI R&D Knowledge Repository**
 - a. **Establish a Unified AI R&D Tracking System** – Develop a database that catalogs all AI-related research projects, funding sources, and implementation outcomes to enhance cross-agency collaboration.
 - b. **Require Agency Participation and Data Sharing** – Mandate that all federal agencies report AI investments, research findings, and technology development progress into the centralized repository to ensure complete visibility.

Expected Impact of These Policies

Implementing these procurement reforms will result in the following:

- **Reduced AI funding duplication** – Streamlined procurement processes will eliminate redundant federal investments, allocating taxpayer dollars efficiently. By improving interagency visibility and coordination, agencies can prioritize high-impact AI initiatives, potentially saving millions in wasted research funding.
- **Faster AI adoption across government** – A fast-track AI procurement process will enable federal agencies to acquire, test, and deploy AI solutions more quickly, allowing them to stay ahead of emerging technological advancements. This will ensure that critical AI capabilities—such as automation, cybersecurity, and data analysis—are integrated into government operations before adversaries and competitors gain an edge.
- **Stronger collaboration between research, industry, and government** – Expanding AI workforce partnerships, commercialization accelerators, and small business incentives will bridge the gap between AI innovators and government agencies. Cross-sector collaboration will enhance knowledge-sharing, create more effective and scalable AI



solutions, and expand opportunities for small businesses to contribute to national AI objectives.

- **More data-driven procurement decisions** – Mandating AI procurement performance metrics will give agencies real-time insights into adoption rates, cost-effectiveness, and operational impact, leading to more strategic and transparent funding decisions. This will increase accountability, prevent misallocation of resources, and ensure long-term sustainability in AI investments.

These reforms will enhance efficiency, reduce costs, and position federal agencies to rapidly adopt AI solutions that drive mission success.

Conclusion

Artificial intelligence is a strategic imperative for the United States, impacting national security, economic growth, and global competitiveness. However, without a robust AI workforce and an optimized procurement process, the nation risks falling behind in the race for AI innovation and deployment. The federal government must take proactive steps to source, recruit, and retain top AI talent while ensuring that AI research and development investments are strategically aligned and efficiently utilized.

This response highlights two critical policy areas that require immediate action:

1. **Education & Workforce Development** – Strengthening AI talent pipelines, improving recruitment and retention efforts, and ensuring hiring practices are merit-based and free from ideological bias.
2. **Procurement & Technology Transition** – Modernizing AI procurement by reducing redundancies, increasing interagency collaboration, and accelerating the transition of AI solutions from research to operational deployment.

Our Commitment to Action

Our organization brings direct experience in tackling these challenges. By implementing the recommended policies, we can address systemic gaps in AI workforce readiness and procurement efficiency, ensuring that the U.S. government:

- **Attracts and retains the best AI talent** to drive innovation.
- **Maximizes the return on AI investments** by avoiding redundant funding.
- **Rapidly deploys AI solutions** to enhance national security and government operations.
- **Maintains a strategic edge in the global AI landscape** through innovative, data-driven policies.

Call to Action



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We urge the Office of Science and Technology Policy (OSTP) and the Networking and Information Technology Research and Development (NITRD) National Coordination Office (NCO) to integrate these recommendations into the AI Action Plan. By prioritizing workforce development and procurement efficiency, the U.S. government can ensure that talent shortages or bureaucratic inefficiencies do not hinder AI innovation.

We look forward to continued engagement and collaboration in shaping a strong, competitive, and ethically grounded AI future for the United States.